Literature Review for Master Thesis

1 Avalanche Modelling

1.1 Numerical investigation of crack propagation regimes in snow fracture experiments (Bobillier et al., 2024)

1.2 Micromechanical modeling of snow snow failure (Bobillier et al., 2020)

1.3 Modeling snow slab avalanches caused by weak-layer failure – Part 2: Coupled mixed-mode criterion for skier-triggered anticracks (Rosendahl & Weißgraeber, 2020)

1.4 Investigating the release and flow of snow avalanches at the slope-scale using a unified model based on the material point method (Gaume et al., 2019)

1.5 Numerical investigation of the mixed-mode failure of snow (Mulak & Gaume, 2019)

1.6 Snow fracture in relation to slab avalanche release: critical state for the onset of crack propagation (Gaume et al., 2017)

1.7 Modeling of crack propagation in weak snowpack layers using the discrete element method (Gaume et al., 2015)

1.8 A physical SNOWPACK model for the Swiss avalanche warning: Part I: numerical model (Bartelt & Lehning, 2002)

1.9 A numerical model to simulate snow-cover stratigraphy for operational avalanche forecasting (Brun et al., 1992)

Main Question of article: How do snow conditions such as temperature gradient and water content influence snow metamporphism and how can this be implemented in the CROCUS-snowpack model?
Why this is relevant for the thesis: Snow metamorphism changes grain properties, which is important for the formation of weak layers which can trigger avalanche release, but also overall stability of the snowpack

When a snowpack is subjected to a change, like a change in temperature or pressure conditions, the snow grains change (Brun et al., 1992). Snow metamorphism changes the grain sizes and shapes, which changes mechanical stability of the snow (Brun et al., 1992). This is relevant for avalanche modelling because certain types of snow metamorphism can weaken layers in snow, such layers can act as failure planes and make release of avalanches easier. Snow metamorphism is affected by the water content of the snow, the capillary pressure of the water between the snow grains, the radius of curvature of the grains (shape), as well as the temperature gradient (Brun et al., 1992).

Brun et al., 1992 describes the main limitation of past methamorphism models as assuming the snowpack to have homogenous shapes and sizes, which are often spherical or simplified geometric shapes, this does not represent snow accurately however, especially fresh snow is very heterogenous. Brun et al., 1992 conducted experiments for different snow types and conditions to determine new equations to describe metamorphism. These experiments showed that the rate and type of metamorphism does not depend on crystal type but rather on the temperature gradient (Brun et al., 1992). A gradient below 5°C/m results in rounded crystals, a gradient above 5°C/m results in facteted crystals, which transform to depth hoar, if a gradient of more than 15°C/m is applied (Brun et al., 1992). This is relevant for avalanche modelling, because it is critical to understand how snow grains evolve to asess their stability, which can be used to determine where weak layers will form. These experiments were conducted for dry and wet snow conditions.

Brun et al., 1992 describe a new description of snow, which takes into account the dendricity, sphericity and size of the grains. When snow falls, new layers with different properties are added to the snowpack (Brun et al., 1992). Simulating these different layers is again important to assess the stability as well as interactions of the different layers. Brun et al., 1992 combined the observations made in the laboratory with the CROCUS snowpack model to simulate these different layers based on meterological observations. Using MEPRA (a system for avalanche risk forecasting), the shear stress of the different layers was estimated at a 45° angle. This is important for avalanche forcasting, because the strength of different layers can be assessed and it can be determined under which stress the layers fail.

1.10 A discrete numerical model for granular assemblies (Cundall & Strack, 1979)

Main Question of article: How can the forces and the resulting displacements be modeled in a granular medium?
Why this is relevant for the thesis: Snow consists of grains, forces at the surface such as mechanical loading can cause displacement of the individual grains. If this happens at a very large scale, it is an avalanche.

The distinct element method described by Cundall and Strack, 1979 is a numerical modeling method to model the mechanical behavior of a material composed of disks spheres. This is relevant to the thesis because snow is composed of individual snow grains, which interact with each other at their contacts, all the grains together form the snow pack. It is important to model the behavior and displacement of the grains to model avalanche dynamics because avalanches form through loading or failure in a weak layer, which then propagates through the entire snowpack (Gaume et al., 2017).

Since a snowpack is a medium composed by individual snow grains, which are not entriely compacted (contain air between the grains) and interact with each other at grain contacts, the medium described by Cundall and Strack, 1979 represents snow well. It is a granular medium composed of distinct particles, which interact with each other at their contacts and form a porous medium (Cundall & Strack, 1979). The loading and unloading of such materials is complex and stresses cannot be measured easily because of the interactions of different grains (Cundall & Strack, 1979).

The distinct elememt method can model particles of any shape (Cundall & Strack, 1979), which is important for snow modelling because the snow grains are not always spherical. Cundall and Strack, 1979 model the transient interaction of particles by calculating the contact forces and displacements of the stressed discs (particles) caused by disturbances at the boundaries. In the avalanche case discussed in the thesis, the distrubance is represented by loading of the surface, which can be caused by a skier, which causes failure in a weak layer, or cracking in the snowpack. This results in stress propagation through the snowpack, which results in displacement of the particles, which is the avalanche motion. The resultant foces on the disks are calculated only by the discs which are in direct contact (all the disks that touch one disk cause its resultant force) (Cundall & Strack, 1979). Cundall and Strack, 1979 describe a model only for dry materials, this means this is only applicable for dry snow avalanches which contain no water between the snow grains.

Numerical modelling, such as described by Cundall and Strack, 1979is advantageus over lab tests and experiments because it is less time consuming and any data can be accessed at any stage, which means that parameters such as loading, particle size, and properties such as stiffness and shape can be changed at any stage. In the avalanche case, this means that simulations can be run for different snow properties and loading scenarios, for example a simulation for older, more compacted snow can be run as well as a simulation for fresh snow. The numerical model for spheres is based on methods described by Serrano and Roderiguez-Ortiz (1973, I CANNOT FIND THIS PAPER ONLINE). The forces at the interfaces between the spheres are calculated by incremental displacements of the centers of particles, where shape changes of the individual particles are neglible (Cundall & Strack, 1979). How can the center of the grain be determined if the grain has an irrgular shape? The matrix representing contact stiffness between the grains described by Cundall and Strack, 1979 needs to be recalculated whenever a new contact is made or broken, meaning when there is a crack in the snowpack or when new contacts between snow grains are made. The calculations of forces are based on Newtons 2nd law, which describes the movement of a particle as the result of the forces acting on it. The deformation is neglected because the deformation of individual particles is small compared to the deformation of the medium as a whole (Cundall & Strack, 1979). This means that the deformation of the individual snow grains is small compared to the deformation of the entire snowpack which represents the avalanche movement. What about the fusing / compaction of grains which changes snow properties? The relative displacement of a particle is calculated by the sum of the force increments acting on it, partciles can also overlap, in this case the contact foces and the interaction forces also need to be calculated (Cundall & Strack, 1979). The location of the rigid boundaries (walls) described by Cundall and Strack, 1979 needs to be specified before modelling. This could be a possible limitation for the avalanche case because there is only one rigid "wall" which is the interface of snow and the rock below it.

2 Crack Propagation

3 Relevant Source Mechanisms

4 Bibliography

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